

Intelligent Drug Sales Forecasting and Healthcare Insights with Artificial Intelligence

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Accurate pharmaceutical sales forecasting can support inventory planning and medicine availability in Sri Lanka. This study developed and evaluated an artificial intelligence-based system for category-wise drug sales forecasting. A hybrid dataset combining sales records from two pharmacies in Kurunegala with synthetic data comprised 517 weekly observations per category across eight pharmaceutical categories from 2014 to 2023. Data preparation included chronological ordering, lag-feature generation, and scaling for selected models. Seven forecasting models, SARIMAX, Prophet, XGBoost, LightGBM, long short-term memory (LSTM), gated recurrent unit (GRU), and Transformer, were compared alongside weighted-average and performance-weighted ensembles. Evaluation used a chronological split with the final ten weeks reserved for testing, measuring mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and inference time. LightGBM achieved the lowest average MAE of 23.3728 and RMSE of 28.9163 across the eight categories. LSTM and GRU achieved average MAEs of 24.0362 and 24.1104, respectively. XGBoost provided the fastest average inference time of 0.0343 seconds. Model performance varied by category, and ensembles did not outperform LightGBM on average MAE or RMSE. A web-based microservice application integrated forecasting, model training, and explainability support. These findings support category-specific model selection for pharmaceutical sales forecasting. However, the hybrid dataset, short test period, limited geographical representation, and absence of external demand variables restrict generalizability. Further evaluation using broader real-world pharmacy data is needed before operational adoption.

Keywords: Drug sales forecasting, Explainable artificial intelligence, LightGBM, Machine learning, Time-series forecasting